THE DETERMINANT FUNCTOR ON PERFECT COMPLEXES OF TOR AMPLITUDE [-1, 0] WITH CANONICAL TRIVIALIZATION SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-quasi-coherence-pullback is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 10550 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Module-level determinant of two-term complexes. More on Algebra, lemma determinant-two-term-complexes (cross-chapter, named). Constructs the invertible module det(L-bullet) over an affine base.
- Quasi-coherence of pullbacks and flat base change. lemma-quasi-coherence-pullback (same chapter, NOT carded, declared external). Underlies the compatibility det(L) tensor B = det(L tensor B) for affine opens V inside U.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08DW: sources/stacks-perfect/pinned-original-source.tex, source lines 581–581. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08E9: sources/stacks-perfect/pinned-original-source.tex, source lines 2508–2508. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0FJM: focused-seeds/geometry/S9-perf2-determinant-two-term-complexes/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FJI: focused-seeds/geometry/S9-perf2-determinant-two-term-complexes/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. There is a functor*

$$\det : \left\{ \begin{array}{l} \text{category of perfect complexes} \\ \text{with tor amplitude in } [-1, 0] \\ \text{morphisms are isomorphisms} \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{category of invertible modules} \\ \text{morphisms are isomorphisms} \end{array} \right\}$$

In addition, given a rank 0 perfect object L of $D(\mathcal{O}_X)$ with tor-amplitude in $[-1, 0]$ there is a canonical element $\delta(L) \in \Gamma(X, \det(L))$ such that for any isomorphism $a : L \to K$ in $D(\mathcal{O}_X)$ we have $\det(a)(\delta(L)) = \delta(K)$. Moreover, the construction is affine locally given by the construction of More on Algebra, Section 0FJI.

Proof. Let L be an object of the left hand side. If $\mathrm{Spec}(A) = U \subset X$ is an affine open, then $L|_U$ corresponds to a perfect complex $L^\bullet$ of A-modules with tor-amplitude in $[-1, 0]$, see Lemmas 06Z0, 08E9, and 08EB. Then we can consider the invertible A-module $\det(L^\bullet)$ constructed in More on Algebra, Lemma 0FJM. If $\mathrm{Spec}(B) = V \subset U$ is another affine open contained in U, then $\det(L^\bullet) \otimes_A B = \det(L^\bullet \otimes_A B)$ and hence this construction is compatible with restriction mappings (see Lemma 08DW and note $A \to B$ is flat). Thus we can glue these invertible modules to obtain an invertible module $\det(L)$ on X. The functoriality and canonical sections are constructed in exactly the same manner. Details omitted. □